23 A uniform copper rod of cross-sectional area 8.0 mm^2 has 8.2×10^{28} conduction electrons per cubic metre. A current flows through the rod when a potential difference of 3.0 V is applied across it.

Given that the drift velocity of electrons in the rod is 2.3×10^{-5} m s^{-1}, what is the resistance of the rod?

A 0.08 Ω

B 0.16 Ω

C 0.80 Ω

D 1.2 Ω